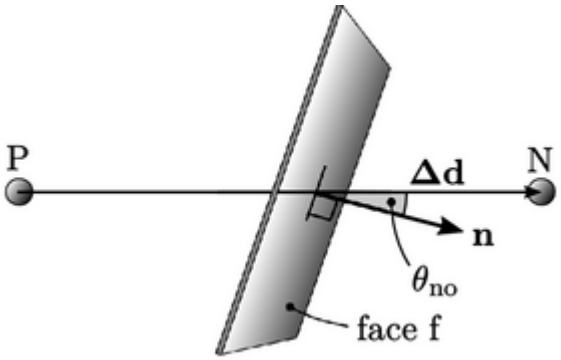
Notes on Computational Fluid Dynamics: General Principles

3.8 Surface normal gradient

The surface normal gradient $\nabla_{\mathbf{n}}\Psi_{\mathrm{f}}$ is a part of the Laplacian discretisation Eq. (3.2), illustrated in the figure below.



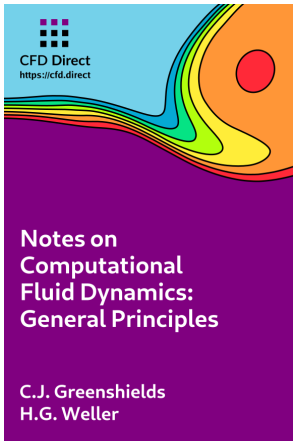
The discretisation of $\nabla_{\mathbf{n}}\Psi_{\mathrm{f}}$ is built upon a finite difference of cell values on each side of the face according to

$$\nabla_{\mathbf{n}}\Psi_{\mathrm{f}} = C_{\Delta}(\Psi_{\mathrm{N}} - \Psi_{\mathrm{P}}), \tag{3.5}$$

where $C_{\Delta} = 1/|\mathbf{\Delta d}|$. When this **orthogonal** scheme is applied to Eq. (3.2) to discretise a Laplacian, it forms coefficients $[\mathbf{A}]$ of a matrix equation $[\mathbf{A}][\mathbf{\Psi}] = [\mathbf{b}]$ since it references cell values of the field $\mathbf{\Psi}$. For cell i, the coefficient for each neighbour cell (N) is $a_{i,\mathrm{N}} = |\mathbf{S}_{\mathrm{f}}|\,\Gamma_{\mathrm{f}} C_{\Delta}$ and the diagonal coefficient is the negative of the sum of neighbour coefficients: $-\sum_{\mathrm{N}} a_{i,\mathrm{N}}$.

Discretisation of $\nabla_{\mathbf{n}}\Psi_{\mathrm{f}}$ by Eq. (3.5) is most accurate when the face is orthogonal to $\mathbf{\Delta d}$, *i.e.* the angle θ_{no} between $\mathbf{n}$ and $\mathbf{\Delta d}$ is zero. However, if the face is *non-orthogonal*, the error associated with Eq. (3.5) increases with θ_{no}.

Non-orthogonal correction



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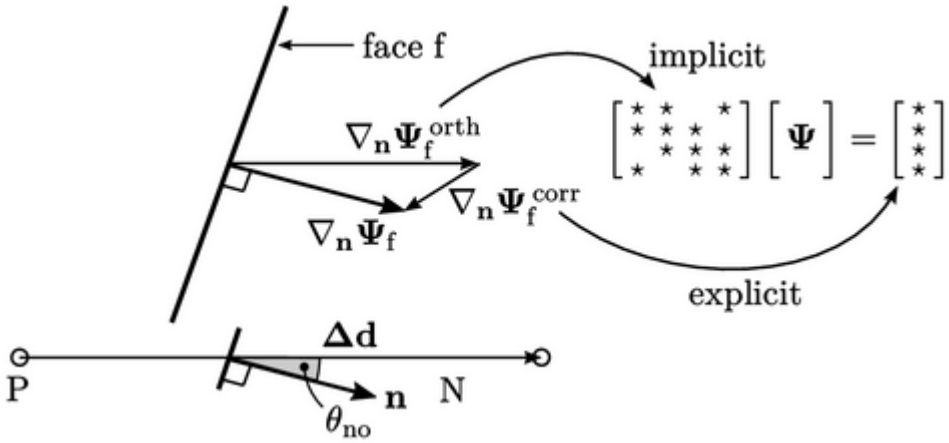
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A more accurate discretisation of $\nabla_\mathbf{n}\Psi_\mathrm{f}$ at a non-orthogonal face is formed of the vector sum of the orthogonal scheme $\nabla_\mathbf{n}\Psi_\mathrm{f}^\mathrm{orth}$ and an *explicit* correction $\nabla_\mathbf{n}\Psi_\mathrm{f}^\mathrm{corr}$. The latter is calculated from the full gradient $\nabla\Psi$ in adjacent cells (described in Sec. 3.15), interpolated to the face $(\nabla\Psi)_\mathrm{f}$.



The correction $\nabla_\mathbf{n}\Psi_\mathrm{f}^\mathrm{corr}$ is explicit, *i.e.* calculated using known values of Ψ, so may need updating within an iterative sequence to maintain accuracy, as discussed in Sec. 5.20. To ensure that the iterative sequence converges, the implicit contribution is elevated by replacing C_Δ in the orthogonal scheme with

$$C_\Delta^\mathrm{corr} = \frac{1}{\mathbf{n}\bullet\Delta\mathbf{d}} = \frac{1}{|\Delta\mathbf{d}|\cos\theta_\mathrm{no}}. \tag{3.6}$$

The **corrected** $\nabla_\mathbf{n}\Psi_\mathrm{f}$ scheme combines the implicit and explicit parts by

$$\nabla_\mathbf{n}\Psi_\mathrm{f} = \underbrace{C_\Delta^\mathrm{corr}(\Psi_\mathrm{N}-\Psi_\mathrm{P})}_{\text{orthogonal, implicit}} + \underbrace{(\mathbf{n}-C_\Delta^\mathrm{corr}\Delta\mathbf{d})\bullet(\nabla\Psi)_\mathrm{f}}_{\text{correction, explicit}}. \tag{3.7}$$

The corrected scheme is generally stable for $\theta_\mathrm{no} \lesssim 75°$. For $\theta_\mathrm{no} > 75°$, stability can be maintained at the expense of accuracy by *limiting* the magnitude of the correction $\nabla_\mathbf{n}\Psi_\mathrm{f}^\mathrm{corr}$ below some fraction of the magnitude of the orthogonal $\nabla_\mathbf{n}\Psi_\mathrm{f}^\mathrm{orth}$ part.

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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